

Adam W. Hansen, PhD

Founder & Chief Executive Officer, Geneial, Inc. | Computational Geneticist

Houston, Texas | LinkedIn | ORCID | Google Scholar

Mendelian genomics | rare disease | patient registries | biomedical AI | interoperability | pharmacogenomics | privacy-preserving computation

EDUCATION AND ADVANCED TRAINING

- 2020–2021** **Texas Medical Center Innovation, Houston, TX** — *Biodesign Fellow, Digital Health*
Full-time interdisciplinary fellowship in clinical need identification, digital health, venture formation, healthcare delivery, and commercialization.
- 2016–2020** **Baylor College of Medicine, Houston, TX** — *PhD, Molecular and Human Genetics*
Human Genome Sequencing Center; advisor: Richard A. Gibbs, PhD. Dissertation: “Accelerating Mendelian Discovery: A Genocentric Approach.” Bioinformatics, Genomics & Systems Biology training track.
- 2009–2016** **Brigham Young University, Provo, UT** — *BS, Bioinformatics*
Minors in Computer Science, Chinese, and Global Business & Literacy.
- 2013** **Nanjing University, Nanjing, China** — *Advanced Chinese Language & Literature*
Study abroad program.

PROFESSIONAL APPOINTMENTS AND LEADERSHIP

- 2020–present** **Geneial, Inc. — Co-founder; Chief Executive Officer (2021–present)**
Founded and leads a precision-medicine software and research organization developing participant-centered registries, electronic consent and longitudinal data collection, genomic/phenotypic interoperability, pharmacogenomic decision support, and AI-assisted biomedical data transformation.
- Principal investigator for approximately \$2.75M in NIH-funded small-business research and PD/PI for a federal R&D subaward supporting NHLBI BioData Catalyst under the ARPA-H Biomedical Data Fabric program.
 - Directed development and evaluation of GenOMA (Geneial Ontology Mapping Agent); the peer-reviewed benchmark received the Best Paper Award — Applications & Practice Track at ML4H 2025.
 - Built multidisciplinary teams spanning computational genetics, biomedical informatics, AI/data science, software engineering, security, product, and research operations; co-inventor on an issued U.S. patent for secure data exchange.
- 2020–2021** **Roxie Health — Co-founder & Board Member**
Digital-health venture formed through the TMC Biodesign Fellowship, focused on data-driven fall-risk identification and prevention for older adults; contributed data-science and strategic support.
- 2016–2020** **Baylor College of Medicine, Human Genome Sequencing Center — Doctoral Researcher**
Developed scalable computational approaches for Mendelian gene discovery and phenotypic expansion. Analyzed exome data from more than 18,000 individuals and identified 154 candidate disease-gene associations; contributed to characterization of POLR2A-, KIF1A-, AHDC1-, DDX3X-, and related neurodevelopmental disorders.
- 2019–2020** **Enventure — ENRICH Program Director**
Directed a biomedical innovation and consulting program connecting multidisciplinary trainee teams with life-science companies and translational projects.
- 2014–2015** **Tute Genomics — Bioinformatics & Project Management Intern**
Led a small team developing a pharmacogenomics annotation and report-generation prototype; supported bioinformatics product development and project coordination. Tute Genomics was later acquired by PierianDx.

EXTRAMURAL SPONSORED RESEARCH AND FEDERAL R&D

- 2024–2025** **Automated Mapping to Phenopackets via a Modular Network of Fine-tuned LLMs** — *PD/PI, Geneial | UNC/RENCI subaward 5132861 | NIH/NHLBI Other Transaction Agreement 1OT3HL147154*
ARPA-H Biomedical Data Fabric / NHLBI BioData Catalyst federal R&D subaward; project period September 1, 2024–August 31, 2025. Led development of agentic tools for transforming understructured clinical and registry data into standards including LinkML and GA4GH Phenopackets.
- 2022–2025** **Privacy-Preserving Connectivity for Rare-Disease Patients** — *Principal Investigator | NIH/NHGRI | R42HG012288 and R44HG012288 (1R42HG012288-01; 4R44HG012288-02)*
Approximately \$2.4M combined Phase I/II Fast-Track STTR/SBIR support. First NHGRI award under NIH's Small Business Transition Grant for Early-Career Scientists; completed NIH I-Corps during the award trajectory.
- 2023–2024** **Increasing the Value of Genomic Medicine through Private Pharmacogenomic Reporting** — *Principal Investigator | NIH/NLM | R43LM014350 (1R43LM014350-01), Phase I SBIR*
\$349,432; project period September 7, 2023–February 29, 2024. Developed and evaluated a privacy-preserving, patient-centered approach for reusing genomic data for pharmacogenomic interpretation and decision support.

INSTITUTIONAL RESEARCH SUPPORT AND FELLOWSHIP FUNDING

- 2018–2020** **Baylor College of Medicine President's Circle Precision Medicine Initiative**
Four team-based competitive awards (two in 2018, one in 2019, and a 2020 extension) supporting genomic discovery and precision-medicine research.
- 2017** **Cullen Foundation Award, Baylor College of Medicine**
- 2016–2017** **NIGMS T32 Training Grant Trainee, Baylor College of Medicine** — *T32GM008307*

PATENTS AND INTELLECTUAL PROPERTY

- 2025** **U.S. Patent 12,417,301 B2 — “Secure data exchange”** — *Inventors: Adam Wesley Hansen and Christopher Glenn Hansen | Assignee: Geneial LLC*
Issued September 16, 2025. U.S. application 18/006,850; publication US20230315875A1. Privacy-preserving systems and methods for secure analysis and exchange across independent data sources.
- Pending** **Related U.S. and international patent applications**
Co-inventor on additional applications covering privacy-preserving data exchange and hardware-accelerated homomorphic encryption, including pending European applications EP21849384.9 and EP22782388.7.

HONORS AND AWARDS

- 2026** Young Alumnus Award, Graduate School of Biomedical Sciences, Baylor College of Medicine
- 2025** Best Paper Award — Applications & Practice Track, Machine Learning for Health Symposium (ML4H 2025)
- 2023** Reviewers' Choice Abstract, American Society of Human Genetics Annual Meeting
- 2019** Reviewers' Choice Abstract, American Society of Human Genetics Annual Meeting (top 10%)
- 2017** Cullen Foundation Award, Baylor College of Medicine
- 2015** Crocker Innovation Fellow, Brigham Young University
- 2015** Second Place, Chinese Speech Competition, Brigham Young University
- 2013–2015** Dean's List, Brigham Young University
- 2012–2013** Cheri de Bonneville Scholarship, Brigham Young University

FEATURED SCHOLARSHIP

Four highlighted works representing senior/corresponding, first-author, or co-first-author contributions across biomedical AI and Mendelian genomics.

1. Di J, Vaughn JR, Proulx J, Nordstrand S, Daines B, Ward KM, Lupo PJ, Hu J, Murugan M, **Hansen AW**. An Agentic Approach to Phenotype Mapping from Rare Disease Surveys. *Proceedings of Machine Learning Research*. 2026;297:1253–1268. *Best Paper Award — Applications & Practice Track, ML4H 2025*.
2. **Hansen AW**, Murugan M, Li H, Khayat MM, Wang L, Rosenfeld JA, et al. A Genocentric Approach to Discovery of Mendelian Disorders. *American Journal of Human Genetics*. 2019;105(5):974–986. doi:10.1016/j.ajhg.2019.09.027. *First author*.
3. **Hansen AW**, Arora P, Khayat MM, Smith LJ, Lewis AM, Rossetti LZ, et al. Germline mutation in POLR2A: a heterogeneous, multi-systemic developmental disorder characterized by transcriptional dysregulation. *Human Genetics and Genomics Advances*. 2021;2(1):100014. doi:10.1016/j.xhgg.2020.100014. *First and corresponding author*.
4. Montenegro-Garreaud X*, **Hansen AW***, Khayat MM, et al. Phenotypic expansion in KIF1A-related dominant disorders: A description of novel variants and review of published cases. *Human Mutation*. 2020;41(12):2094–2104. doi:10.1002/humu.24118. **Co-first authors; Hansen co-corresponding author*.

ADDITIONAL PEER-REVIEWED PUBLICATIONS

5. Silva PJ, Rahimzadeh V, Powell R, Husain J, Grossman S, **Hansen A**, Hinkel J, Rosengarten R, Ory MG, Ramos KS. Health equity innovation in precision medicine: data stewardship and agency to expand representation in clinicogenomics. *Health Research Policy and Systems*. 2024;22:170. doi:10.1186/s12961-024-01258-9. *Peer-reviewed Comment; developed from a Texas A&M Health clinicogenomics/health-equity symposium*.
6. Baxter SM, Posey JE, Lake NJ, et al.; **Centers for Mendelian Genomics Consortium**. Centers for Mendelian Genomics: A decade of facilitating gene discovery. *Genetics in Medicine*. 2022;24(4):784–797. doi:10.1016/j.gim.2021.12.005. *Consortium publication listed on Google Scholar/ResearchGate profile*.
7. Khayat MM, Hu J, Jiang Y, Li H, Chander V, Dawood M, **Hansen AW**, et al. AHDC1 missense mutations in Xia-Gibbs syndrome. *Human Genetics and Genomics Advances*. 2021;2(4):100049. doi:10.1016/j.xhgg.2021.100049.
8. Brownstein CA, Smith RS, Rodan LH, et al.; **Hansen AW**, et al. RCL1 copy number variants are associated with a range of neuropsychiatric phenotypes. *Molecular Psychiatry*. 2021;26(5):1706–1718. doi:10.1038/s41380-021-01035-y.
9. Khayat MM, Li H, Chander V, Hu J, **Hansen AW**, et al. Phenotypic and protein localization heterogeneity associated with AHDC1 pathogenic protein-truncating alleles in Xia-Gibbs syndrome. *Human Mutation*. 2021;42(5):577–591. doi:10.1002/humu.24190.
10. Wang X, Posey JE, Rosenfeld JA, et al.; **Hansen AW**, et al. Phenotypic expansion in DDX3X — a common cause of intellectual disability in females. *Annals of Clinical and Translational Neurology*. 2018;5(10):1277–1285. doi:10.1002/acn3.622.
11. Pope WH, Bowman CA, Russell DA, et al.; SEA-PHAGES; Phage Hunters Integrating Research and Education (PHIRE); Mycobacterial Genetics Course. Whole genome comparison of a large collection of mycobacteriophages reveals a continuum of phage genetic diversity. *eLife*. 2015;4:e06416. doi:10.7554/eLife.06416. *Consortium publication; Hansen AW listed as a SEA-PHAGES/PHIRE program member*.

CONFERENCE ABSTRACTS AND POSTERS

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| 2026 | Nordstrand S, Kamauu A, Flinders J, Hansen A. “Advancing Computable Operational Definitions with Ontology Mapping and Synthetic Data Simulation for More Consistent Real-World Evidence.” ISPOR 2026, Philadelphia, PA. Poster RWD139. |
| 2026 | Nordstrand S, Kamauu A, Flinders J, Hansen AW. “Advancing Computable Operational Definitions with Ontology Mapping and Synthetic Data Simulation for More Consistent Real-World Evidence.” 42nd Annual Meeting of the International Society for Pharmacoepidemiology (ICPE/ISPE), Milan, Italy. Poster B-268. |
| 2025 | Shapiro D, Jodon Z, Ward K, Ward K, Hansen CG, Hansen AW. “Enabling Privacy-Preserving Clinical Decision Support for Pharmacogenomics from Whole Genome Sequencing.” American Society of Human Genetics Annual Meeting. Poster 5088T. |
| 2025 | Proulx J, Daines B, Vaughn J, Nordstrand S, Hansen AW. “An Agentic AI Tool for Automated Data Transformation from TOPMed to LinkML and Phenopackets.” American Society of Human Genetics Annual Meeting. Poster 2019F. |

2025	Vaughn J, Di J, Proulx J, Nordstrand S, Daines B, Lupo PJ, Hu J, Hansen AW. "Automating HPO Term Extraction from Clinical Text Using an LLM Pipeline." American Society of Human Genetics Annual Meeting. Poster 2032F.
2025	Daines B*, Vaughn J*, Proulx J, Hu J, Gerber T, Gibbs RA, Hansen AW. "An AI Co-scientist Framework for Reproducible Discovery in Rare Disease Registries." American Society of Human Genetics Annual Meeting. Poster 2031F.
2025	Nordstrand S, Ward K, Sharma Chapagain B, Wangler MF, Hansen AW. "Low-cost, Interoperable Rare Disease Registries via an Enhanced GenomeConnect Survey." American Society of Human Genetics Annual Meeting. Poster 7018F.
2025	Gillentine M, Aigner S, et al.; Hansen A, et al. "Damaging Variants in HNRNPUL2 Result in a Novel HNRNP-Related Neurodevelopmental Disorder." American Society of Human Genetics Annual Meeting. Poster 7047W.
2025	Gillentine MA, Margolis L, Hansen A, Aigner S, Ward K, et al. "Understanding the Autism Phenotype Among Individuals with HNRNP-Related Neurodevelopmental Disorders." International Society for Autism Research (INSAR) Annual Meeting.
2025	Gillentine M, Hansen A, Da Silva C, Phillips L, Plecas Maricic S, et al. "HNRNP-Related Neurodevelopmental Disorders: Expanding Knowledge of Known Conditions and Prompting Novel Studies into Candidate Genes." American College of Medical Genetics and Genomics (ACMG) Annual Clinical Genetics Meeting. Poster P263.
2024	Hansen A, Chinapuvvula S, Rideau T-T, Ward K, Hansen C, Wang LL. "Characterization of Hearing Loss and Skin Cancer in Individuals with Rothmund-Thomson Syndrome." American Society of Human Genetics Annual Meeting. Poster 2019F.
2024	Shapiro D, Belliston D, Al Rabadi R, Greer T, Jodon Z, Ward K, Hansen C, Hansen A. "Using Whole Genome Sequencing to Call Star Alleles Expands Coverage and Sensitivity for Clinical Pharmacogenomics Decision-Making." American Society of Human Genetics Annual Meeting. Poster 7023F.
2024	Hu J, Hansen A, Ward K, Murugan M, Dawood M, Posey JE, Lupski JR, Wangler M, Gibbs RA. "Haploinsufficiency versus Gain of Function in AHDC1 and the Xia-Gibbs Syndrome." American Society of Human Genetics Annual Meeting. Poster 6041F.
2024	Ward K, Hansen A, Hansen C, Kuldell J, Kaplan C. "Establishing a Comprehensive Registry and Natural History Study for Pol II-Associated Developmental Disorders." American Society of Human Genetics Annual Meeting. Poster 2032W.
2024	Gillentine M, Hansen A, Da Silva C, Phillips L, Plecas Maricic S, et al. "HNRNP-Related Neurodevelopmental Disorders: Expanding Knowledge of Known Conditions and Prompting Novel Studies into Candidate Genes." American Society of Human Genetics Annual Meeting. Poster 2045T.
2023	Hansen A, Ward K, Belliston D, Greer TR, Gerber T, Hansen CG, Murugan M, Hu J, Gibbs RA. "Advancing Insights in Xia-Gibbs Syndrome: A Universal Registry Platform for Phenotypic Analysis and Longitudinal Patient Engagement." American Society of Human Genetics Annual Meeting. Poster PB4624. Reviewers' Choice Abstract.
2022	Hansen A, Little E, Belliston D, Murugan M, Hu J, Hansen C, Gibbs R. "A Private, Universal Registry for Patient Engagement in Biomedical Research: Xia-Gibbs Syndrome as a Model." American Society of Human Genetics Annual Meeting. Poster PB3193.
2019	Hansen AW, et al. "A Genocentric, Optimization-based Approach to Discovery of Mendelian Disease Genes." American Society of Human Genetics Annual Meeting, Houston, TX. Program no. 1393. Reviewers' Choice Abstract.
2019	Hansen AW, et al. "A Genocentric, Optimization-based Approach to Discovery of Mendelian Disease Genes." Cold Spring Harbor Laboratory Biology of Genomes Meeting, Cold Spring Harbor, NY.
2018	Hansen AW, et al. "A Hadoop Data Lake Enables Genocentric Mining for Mendelian Disease Discovery." American Society of Human Genetics Annual Meeting, San Diego, CA.
2015	Hansen AW, et al. Smartphone-based snore and sleep-apnea detection research. Biotechnology & Bioinformatics Symposium, Brigham Young University.
2010	Hansen AW, et al. Characterization of the mycobacteriophage PhrostyMug. American Society for Microbiology Intermountain Branch Meeting.

SELECTED INVITED PRESENTATIONS, PANELS, AND MODERATION

2025	Panelist, "Precision Medicine in MedTech: Shaping the Future of Personalised Care," MedTech Malta / MedTech World, Malta, November 14, 2025.
2025	Invited moderator, Data Panel Conversation, Pediatric Mental Health Summit, Austin, TX, June 6, 2025.

2025	Oral presentation co-author, “An Agentic Approach to Phenotype Mapping from Rare Disease Surveys,” Machine Learning for Health Symposium (ML4H 2025), San Diego, CA.
2024	Invited presentation, “The Xia-Gibbs Syndrome Registry App and Platform: Geneial,” Xia-Gibbs Syndrome Family Conference.
2023	Panelist, Health Equity Innovation Summit 3.0: “Patient-Centric Data Governance and Precision Medicine for All,” Texas A&M Health, Houston, TX, November 2023.
2023	Panelist, “Start-ups: Where to Start?,” Baylor College of Medicine Graduate School Career Series, Houston, TX, September 13, 2023.
2023	Invited speaker/panelist, A.I. Healthcare Transformation Summit, Health Tech Austin, Austin, TX.
2020	“Accelerating Mendelian Discovery: A Genocentric Approach.” NHGRI Genome Sequencing Program Annual Meeting.
2019	“A Genocentric, Optimization-based Approach to Discovery of Mendelian Disease Genes.” NHGRI Genome Sequencing Program Annual Meeting.

SCIENTIFIC PEER REVIEW AND PROFESSIONAL SERVICE

2024–present	National Institutes of Health — Invited Grant Reviewer Invited reviewer for NIH small-business, genomics, informatics, and health-services study sections / special emphasis panels. • Small Business Special Emphasis Panel — ZRG1 MCST-S (14), July 2025 • Small Business Informatics Tools for the Pangenome — MCST (16), July 2025 • Genome Research Study Section — GNOM-G, April 2025 • Small Business: Health Services and Systems B — ZRG1 HSS-K (11), January 2025 • Small Business: Health Services and Systems B — ZRG1 HSS-K (10), December 2024
2021–2022	Brain & Development — Ad hoc manuscript reviewer Completed three peer reviews between August 2021 and January 2022; recognized by the journal’s editors for review contributions.
Current	University of Utah Department of Biomedical Informatics — Industry Advisory Board Member

TEACHING AND MENTORING

2021–present	Research and technical mentorship, Geneial Mentors trainees and early-career staff in bioinformatics, computational genetics, biomedical AI, research design, grant development, and scientific communication.
2022–present	Volunteer English instructor, Sugar Land, TX Weekly English-language instruction for adult learners through a community education program.
2014	Teaching Assistant, Chinese 102, Brigham Young University
2012–2013	Mandarin Chinese Instructor, Missionary Training Center, Provo, UT

RESEARCH TRAINING AND EARLIER RESEARCH EXPERIENCE

2017	Human Genome Sequencing Center, Baylor College of Medicine — Richard A. Gibbs laboratory Transitioned from rotation to doctoral research in large-scale Mendelian disease discovery, data lakes, and phenotypic expansion.
2016–2017	Baylor College of Medicine — Graduate research rotations Research rotations with Huda Zoghbi, MD; Sharon Plon, MD, PhD; Rui Chen, PhD; and Richard A. Gibbs, PhD, spanning neurogenetics, cancer genetics/genomic medicine, computational genomics, and Mendelian disease discovery.
2014–2016	Brigham Young University — Stephen Piccolo Computational Biology Laboratory Undergraduate research in computational biology, reproducible analysis, and biomedical data science.
2015	Brigham Young University — Genetics of Human Disease research course (John S.K. Kauwe, PhD) Applied genome-wide association and statistical genetics approaches to human-disease questions.
2009–2010	Brigham Young University — Phage Hunters Integrating Research and Education (PHIRE) Isolated and characterized mycobacteriophage PhrostyMug; contributed to the broader SEA-PHAGES research effort.

RESEARCH SOFTWARE, DATA PRODUCTS, AND OPEN RESEARCH OUTPUTS

Open-source research software developed through Geneial federal R&D programs, including the ARPA-H Biomedical Data Fabric / NHLBI BioData Catalyst project.

2026	GenOMA (Geneial Ontology Mapping Agent) — open-source agentic ontology-mapping framework for mapping patient-reported rare-disease survey text to Human Phenotype Ontology terms; peer-reviewed benchmark published in PMLR. Public repository: github.com/geneialco/GenOMA-Geneial-Ontology-Mapping-Agent .
2025	McMurry J, Sizer A, Allaway R, et al.; Hansen A, et al. “Critical Bottlenecks in Rare Disease Research and Care: A Community Perspective.” Community white paper, Zenodo. doi: 10.5281/zenodo.14907384.
2024–2025	Geneial Transform Agent — open-source AI agent for transforming understructured biomedical data, including TOPMed-style inputs, into LinkML or GA4GH Phenopackets, with automated schema generation and validation. Public repository: github.com/geneialco/geneial-transform-agent .
2024–2025	UMLS Server — open-source terminology infrastructure and API for fast Unified Medical Language System concept lookup, ontology access, and biomedical NLP workflows. Public repository: github.com/geneialco/umls-server .
2024–2025	synTOPIa Data Generator — open-source toolkit for synthetic phenotype data generation, including workflows for synthesizing TOPMed/dbGaP-style data for development and testing. Public repository: github.com/geneialco/synTOPIa-data-gen .
2013	Mycobacteriophage PhrostyMug complete genome sequence — GenBank accession KF279415.1.
2012	Mycobacteriophage KLUcky39 sequence record — GenBank accession JF704099.1.

HUMAN-SUBJECTS AND RESEARCH COMPLIANCE TRAINING

2026	HHS Office for Human Research Protections — Human Research Protection Foundational Training Completed all five foundational lessons; completion certificates available for IRB and sponsor submissions.
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RESEARCH AND TECHNICAL EXPERTISE

Mendelian disease discovery; exome and genome analysis; rare-disease natural-history registries; patient-reported outcomes; phenotype ontologies (HPO/UMLS); biomedical data interoperability; LinkML and GA4GH Phenopackets; pharmacogenomics; AI/LLM agents; research data harmonization; privacy-enhancing technologies and homomorphic encryption; FAIR data principles; research software/product translation; Python, R, SQL, cloud and large-scale data workflows.